Bow Valley College

Course Outline

MAT1791

Mathematics 10 C
Winter 2021 - Fall 2024

Last Updated: 11/17/2020 4:21:29 PM

Care has been taken to obtain copyright permission to reproduce this material. Any information that will enable Bow Valley College to obtain copyright clearance for any material not acknowledged would gladly be received by:

Bow Valley College
345 6th Avenue SE
Calgary AB T2G 4V1
Attn: Copyright Officer
email: copyright@bowvalleycollege.ca

© Bow Valley College

MAT1791 Mathematics 10 C

COURSE DESCRIPTION

This is an Alberta Education credit course. Topics include spatial sense and reasoning, algebra and number sense, graphing, and study of relations.

REQUISITES	Earn a minimum grade of C+ in each of the following courses: • MATH0801 - Prep Math 10 (5)
EQUIVALENTS	None
CREDITS	5
HOURS	100
ELIGIBLE FOR PLAR	No
ZERO TEXTBOOK COST	

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Develop and apply the primary trigonometric ratios (sine, cosine, tangent) to solve problems that involve right triangles.	2, 3, 5, 7
2	Solve problems that involve linear measurement, using SI and imperial units of measure, estimation strategies and measurement strategies.	2, 3, 5, 7
3	Apply proportional reasoning to problems that involve conversions between SI and Imperial units of measure.	2, 3, 5, 7
4	Solve problems, using SI and imperial units that involve the surface area and volume of 3-D objects.	2, 3, 5, 7
5	Demonstrate an understanding of factors of whole numbers by determining the prime factors, greatest common factor, least common multiple, square root and cube root.	2, 3, 5, 7
6	Demonstrate an understanding of irrational numbers .	2, 3, 5, 7
7	Demonstrate an understanding of powers with integral and rational exponents.	2, 3, 5, 7
8	Demonstrate an understanding of the multiplication of polynomial expressions concretely, pictorially and symbolically	2, 3, 5, 7
9	Demonstrate an understanding of common factors and trinomial factoring, concretely, pictorially and symbolically.	2, 3, 5, 7
10	Interpret and explain the relationships among data, graphs and situations.	2, 3, 5, 7
11	Demonstrate an understanding of relations and functions.	2, 3, 5, 7
12	Demonstrate an understanding of slope.	2, 3, 5, 7
13	Describe and represent linear relations.	2, 3, 5, 7
14	Determine the characteristics of the graphs of linear relations.	2, 3, 5, 7
15	Relate linear relations expressed in slope-intercept, general and slope-point forms.	2, 3, 5, 7
16	Determine the equation of a linear relation.	2, 3, 5, 7
17	Represent a linear function, using function notation.	2, 3, 5, 7
18	Solve problems that involve systems of linear equations in two variables, graphically and algebraically.	2, 3, 5, 7

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Weeks 1, 2 & 3	Numbers and Exponents
Weeks 4, 5 & 6	Measurement and Trigonometry
Weeks 7, 8 & 9	Polynomial Operations; Factoring Polynomial Expressions
Weeks 10, 11 & 12	Relations and Functions; Characteristics of Linear Relations
Weeks 13, 14 & 15	Equations of Linear Relations; Systems of Linear Equations

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1 - 18	Assignments	25%
1 - 18	Tests	75%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A minimum grade of D is required to pass this course. However, a program may require a higher grade in this course to progress in the program or to meet specific program completion requirements.

A grade of C is required to take MAT2791; a grade of C- is required to take MAT2792.

Please consult with the program area or contact the program chair for further details. A minimum Grade Point Average of 2.0 is required for graduation.

GRADING SCHEME

Grade	**Percentage**	**Grade Point**	**Description**
A+	95-100	4.0	Exceptional: superior knowledge of subject matter
A	90-94	4.0	Excellent: outstanding knowledge of subject matter

A-	85-89	3.67	
B+	80-84	3.33	
B	75-79	3.0	Very Good: knowledge of subject matter generally mastered
B-	70-74	2.67	
C+	67-69	2.33	
C	64-66	2.0	Satisfactory/Acceptable: knowledge of subject matter adequately mastered
C-	60-63	1.67	
D+	57-59	1.33	
D	50-56	1.0	Minimal Pass
F	Less than 50	0.0	Fail: an unsatisfactory performance

REQUIRED LEARNING RESOURCES

Textbook and additional learning resources will be found in the Course Offering Information and in Brightspace.

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

The lowest test mark will be replaced by the average of the other test marks.

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To

make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.